Week 1 Assignment

Python Fundamentals for Data Analytics

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Branch	B.E. Computer Science & Engineering (IoT, Cybersecurity and Blockchain)
Duration	1 Week
Total Marks	100

Contents

Task 1: Python Basics
Task 2: Variables & Data Types
Task 3: Operators
Task 4: Conditional Statements
Task 5: Loops
Task 6: Functions
Task 7: Strings & Collections
Task 8: Basic File Handling
Task 9: Mini Project: Student Record Management System

Task 1: Python Basics

Explanation

This program prints a welcome message and then asks for my name, college name and branch using input(). After that it prints all three details in a neat formatted block.

Source code (task1_basics.py)

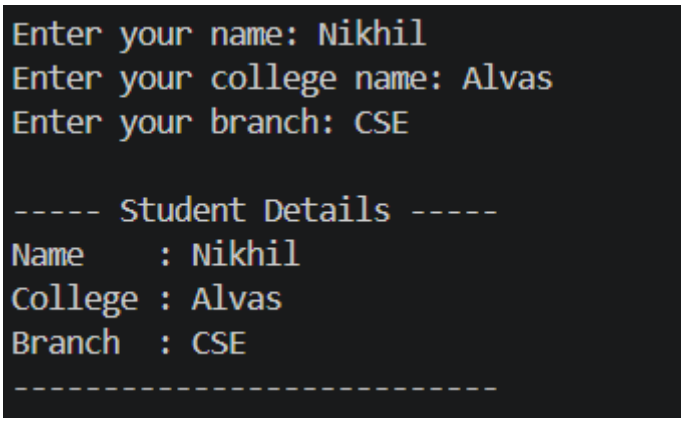
```
# Task 1: Python Basics

print("Welcome to my Python program!")
print()

name = input("Enter your name: ")
college = input("Enter your college name: ")
branch = input("Enter your branch: ")

print()
print("----- Student Details -----")
print("Name    :", name)
print("College :", college)
print("Branch  :", branch)
print("---------------------------")
```

Output



Task 2: Variables & Data Types

Explanation

Here I made four variables of different types: an integer (age), a float (height), a string (city) and a boolean (is_student). Each one is printed together with its type using the type() function.

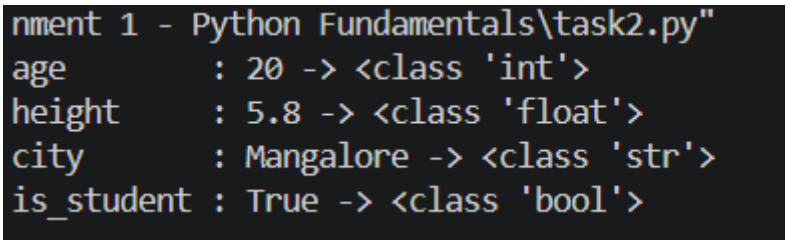
Source code (task2_datatypes.py)

```
# Task 2: Variables and Data Types

age = 20                # integer
height = 5.8            # float
city = "Mangalore"      # string
is_student = True       # boolean

print("age        :", age, "->", type(age))
print("height     :", height, "->", type(height))
print("city       :", city, "->", type(city))
print("is_student :", is_student, "->", type(is_student))
```

Output



Task 3: Operators

Explanation

The calculator takes two numbers from the user and prints the result of addition, subtraction, multiplication, division and modulus. I added a small check so the program does not crash when the second number is 0.

Source code (task3_calculator.py)

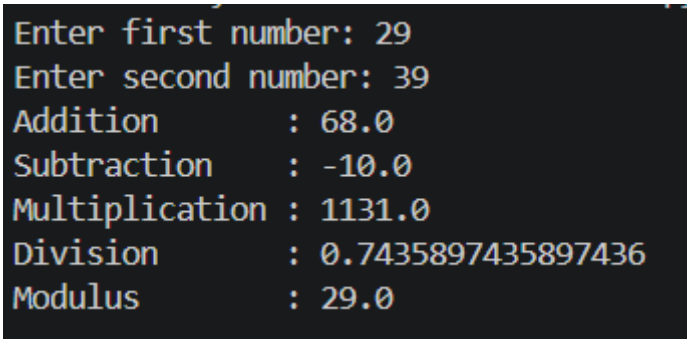
```
# Task 3: Simple Calculator

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Addition       :", num1 + num2)
print("Subtraction    :", num1 - num2)
print("Multiplication :", num1 * num2)

# division and modulus fail if the second number is 0
if num2 == 0:
    print("Division       : Cannot divide by zero")
    print("Modulus        : Cannot divide by zero")
else:
    print("Division       :", num1 / num2)
    print("Modulus        :", num1 % num2)
```

Output



Task 4: Conditional Statements

Explanation

The program reads the marks and uses if, elif and else to decide the grade: 90 and above is A, 75 to 89 is B, 60 to 74 is C, and anything below 60 is Fail.

Source code (task4_grades.py)

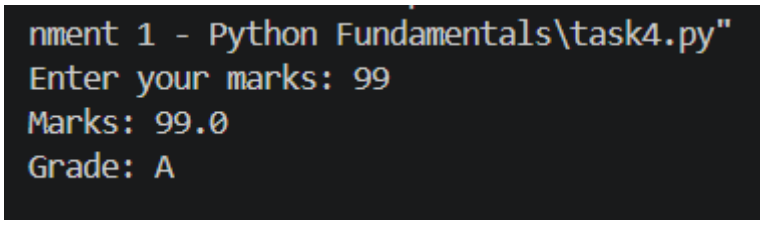
```
# Task 4: Grade Calculator

marks = float(input("Enter your marks: "))

if marks >= 90:
    grade = "A"
elif marks >= 75:
    grade = "B"
elif marks >= 60:
    grade = "C"
else:
    grade = "Fail"

print("Marks:", marks)
print("Grade:", grade)
```

Output



Task 5: Loops

Explanation

This file has three parts. A for loop prints numbers from 1 to 20. Another for loop prints the multiplication table of a number given by the user. A while loop goes from 1 to 50 and prints only the even numbers using the % operator.

Source code (task5_loops.py)

```
# Task 5: Loops

# 1. numbers from 1 to 20 using for loop
print("Numbers 1 to 20:")
for i in range(1, 21):
    print(i, end=" ")
print()

# 2. multiplication table
n = int(input("\nEnter a number for its table: "))
print("Multiplication table of", n)
for i in range(1, 11):
    print(n, "x", i, "=", n * i)

# 3. even numbers from 1 to 50 using while loop
print("\nEven numbers 1 to 50:")
num = 1
while num <= 50:
    if num % 2 == 0:
        print(num, end=" ")
    num += 1
print()
```

Output

```
nment 1 - Python Fundamentals\task5.py"
Numbers 1 to 20:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Enter a number for its table: 21
Multiplication table of 21
21 x 1 = 21
21 x 2 = 42
21 x 3 = 63
21 x 4 = 84
21 x 5 = 105
21 x 6 = 126
21 x 7 = 147
21 x 8 = 168
21 x 9 = 189
21 x 10 = 210

Even numbers 1 to 50:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
```

Task 6: Functions

Explanation

I created two functions. square() returns the square of a number and average() returns the average of three numbers. Both are called with values typed in by the user.

Source code (task6_functions.py)

```
# Task 6: Functions

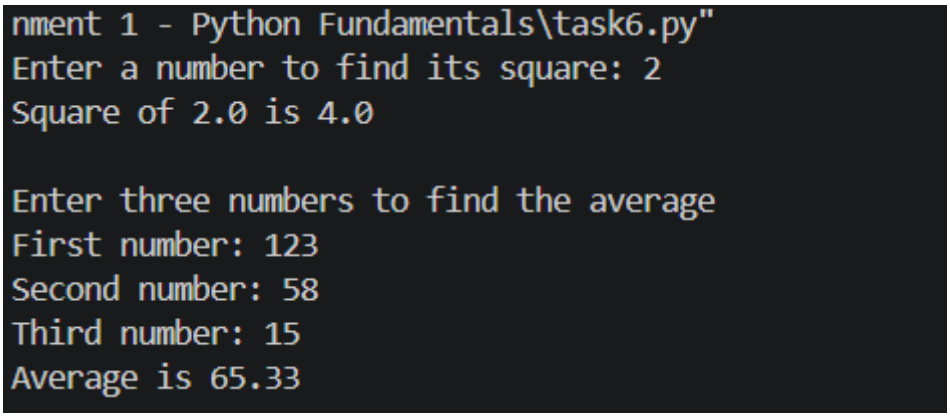
def square(x):
    return x * x


def average(a, b, c):
    return (a + b + c) / 3


n = float(input("Enter a number to find its square: "))
print("Square of", n, "is", square(n))

print()
print("Enter three numbers to find the average")
x = float(input("First number: "))
y = float(input("Second number: "))
z = float(input("Third number: "))
print("Average is", round(average(x, y, z), 2))
```

Output



Task 7: Strings & Collections

Explanation

This program shows string methods (upper, lower, replace, find), list methods (append, remove, sort), a tuple with indexing, a dictionary that stores student information, and a set with add and remove.

Source code (task7_collections.py)

```
# Task 7: Strings and Collections

# ---- Strings ----
text = "Python for Data Analytics"
print("Original :", text)
print("upper()  :", text.upper())
print("lower()  :", text.lower())
print("replace():", text.replace("Data Analytics", "Beginners"))
print("find()   :", text.find("Data"))   # index where "Data" starts

# ---- Lists ----
print("\n--- List ---")
fruits = ["mango", "apple", "banana"]
print("Start    :", fruits)
fruits.append("orange")
print("append   :", fruits)
fruits.remove("apple")
print("remove   :", fruits)
fruits.sort()
print("sort     :", fruits)

# ---- Tuple ----
print("\n--- Tuple ---")
days = ("Mon", "Tue", "Wed", "Thu", "Fri")
print("Tuple    :", days)
print("days[0]  :", days[0])
print("days[-1] :", days[-1])

# ---- Dictionary ----
print("\n--- Dictionary ---")
student = {
    "name": "Nikhil",
    "branch": "CSE",
    "year": 3,
    "cgpa": 8.2
}
print(student)
print("Name:", student["name"])

# ---- Set ----
print("\n--- Set ---")
skills = {"Python", "SQL", "Excel"}
print("Start    :", skills)
skills.add("Power BI")
print("add      :", skills)
skills.remove("Excel")
print("remove   :", skills)
```

Output

```
nment 1 - Python Fundamentals\task7.py"
Original : Python for Data Analytics
upper()  : PYTHON FOR DATA ANALYTICS
lower()  : python for data analytics
replace(): Python for Beginners
find()   : 11

--- List ---
Start    : ['mango', 'apple', 'banana']
append   : ['mango', 'apple', 'banana', 'orange']
remove   : ['mango', 'banana', 'orange']
sort     : ['banana', 'mango', 'orange']

--- Tuple ---
Tuple    : ('Mon', 'Tue', 'Wed', 'Thu', 'Fri')
days[0]  : Mon
days[-1] : Fri

--- Dictionary ---
{'name': 'Nikhil', 'branch': 'CSE', 'year': 3, 'cgpa': 8.2}
Name: Nikhil

--- Set ---
Start    : {'Python', 'Excel', 'SQL'}
add      : {'Python', 'Excel', 'Power BI', 'SQL'}
remove   : {'Python', 'Power BI', 'SQL'}
```

Task 8: Basic File Handling

Explanation

The program creates a text file called introduction.txt, writes a short introduction into it, closes it, then opens it again in read mode and prints what was written.

Source code (task8_filehandling.py)

```
# Task 8: Basic File Handling

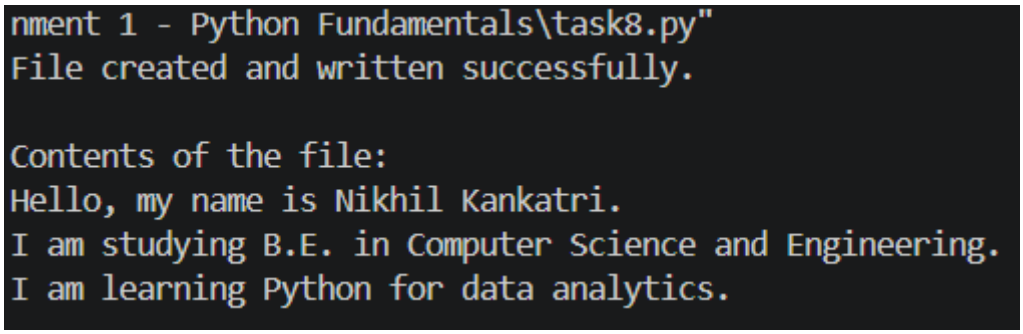
filename = "introduction.txt"

# create the file and write my introduction
f = open(filename, "w")
f.write("Hello, my name is Nikhil Kankatri.\n")
f.write("I am studying B.E. in Computer Science and Engineering.\n")
f.write("I am learning Python for data analytics.\n")
f.close()
print("File created and written successfully.\n")

# read it back and display
f = open(filename, "r")
content = f.read()
f.close()

print("Contents of the file:")
print(content)
```

Output



Task 9: Mini Project: Student Record Management System

Explanation

This is a menu-driven program. Student records are stored as dictionaries inside a list. The menu lets the user add a student, display all students, search a student by name and delete a student by roll number. It also stops the same roll number from being added twice. The menu keeps repeating in a while loop until the user chooses Exit.

Source code (task9_student_records.py)

```
# Task 9: Student Record Management System
# Records are stored in a list of dictionaries

students = []


def add_student():
    roll = input("Enter roll number: ")
    # don't allow the same roll number twice
    for s in students:
        if s["roll"] == roll:
            print("A student with this roll number already exists.")
            return
    name = input("Enter name: ")
    branch = input("Enter branch: ")
    marks = input("Enter marks: ")
    students.append({"roll": roll, "name": name, "branch": branch, "marks": marks})
    print("Student added.")


def display_students():
    if len(students) == 0:
        print("No records found.")
        return
    print("\nRoll\tName\t\tBranch\tMarks")
    print("-" * 40)
    for s in students:
        print(s["roll"] + "\t" + s["name"] + "\t\t" + s["branch"] + "\t" + s["marks"])

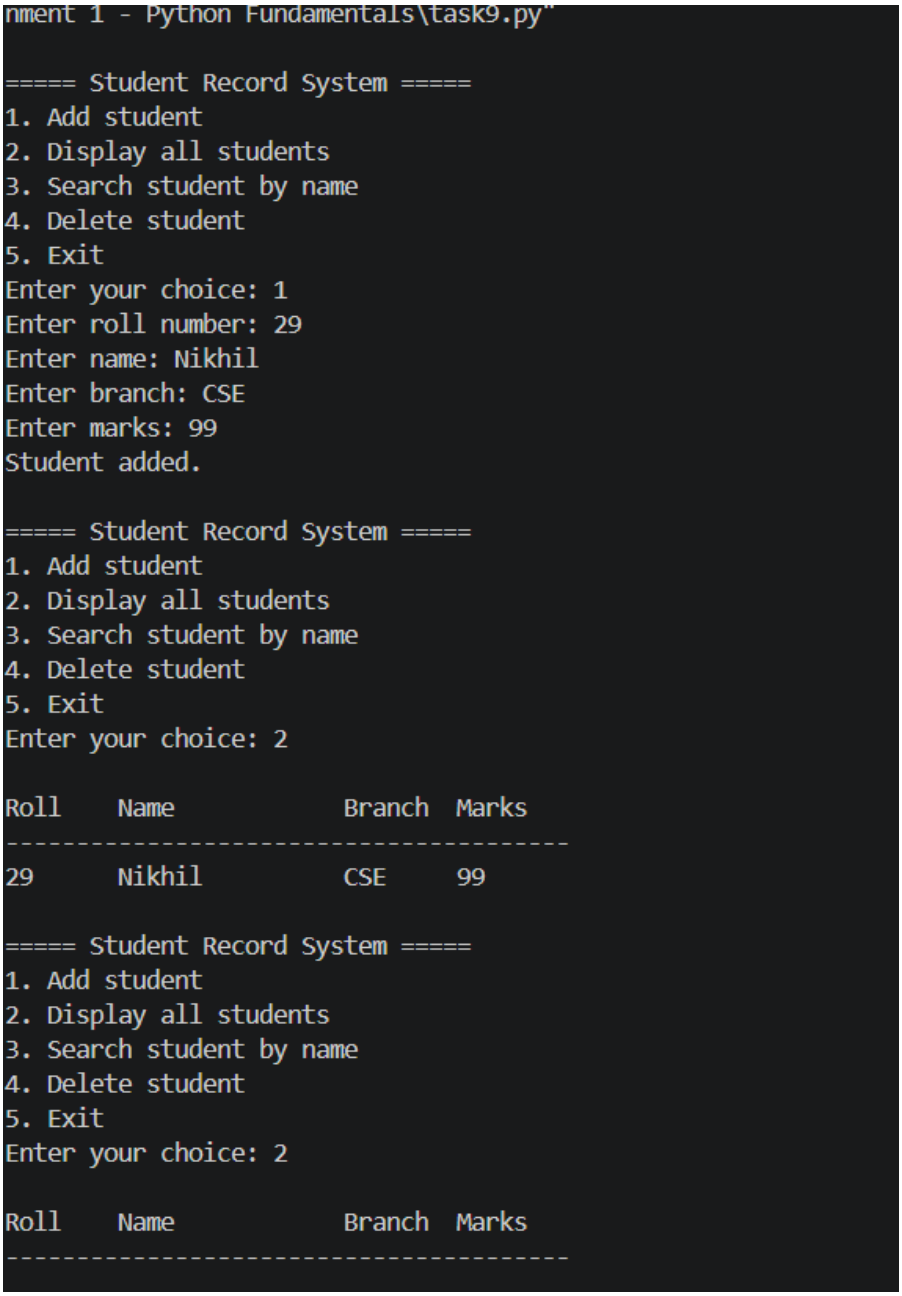

def search_student():
    name = input("Enter name to search: ").lower()
    found = False
    for s in students:
        if s["name"].lower() == name:
            print("Found ->", s["roll"], s["name"], s["branch"], s["marks"])
            found = True
    if not found:
        print("Student not found.")


def delete_student():
    roll = input("Enter roll number to delete: ")
    for s in students:
        if s["roll"] == roll:
            students.remove(s)
            print("Record deleted.")
            return
    print("No student with that roll number.")


while True:
    print("\n===== Student Record System =====")
    print("1. Add student")
    print("2. Display all students")
    print("3. Search student by name")
    print("4. Delete student")
    print("5. Exit")
    choice = input("Enter your choice: ")
```

```
if choice == "1":
    add_student()
elif choice == "2":
    display_students()
elif choice == "3":
    search_student()
elif choice == "4":
    delete_student()
elif choice == "5":
    print("Goodbye!")
    break
else:
    print("Invalid choice, try again.")
```

Output



```
Roll    Name            Branch  Marks
----------------------------------------
29      Nikhil          CSE     99

===== Student Record System =====
1. Add student
2. Display all students
3. Search student by name
4. Delete student
5. Exit
Enter your choice: 5
Goodbye!
```